N-Way vs. Recursive Bisection: A General Architectural Comparison

B.5 — Algorithm Track

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Abstract

Congressional redistricting algorithms divide into two broad families: those that partition a state graph directly into k parts (*n-way*), and those that recursively bisect into two halves until k leaves are reached (*recursive bisection*). Prior work (B.3, B.4) established a small compactness advantage for recursive bisection under edge weighting for congressional chambers. This paper generalises to 450 chamber-year combinations — congressional, state senate, and state house — across all 50 states at census-tract and block-group resolution. We find that recursive bisection outperforms n-way by +0.003–+0.004 mean Polsby–Popper across all chamber types (paired t-test, $p < 0.001$), while n-way is faster by 18–34% only for state house chambers with $k > 80$ — an absolute saving under 200 ms. Prime-k chambers exhibit a narrowed advantage, recoverable via post-processing. For redistricting purposes, recursive bisection is the dominant strategy: the compactness advantage is consistent and statistically significant, and the runtime difference is negligible on modern hardware.

1 Introduction

Congressional redistricting requires partitioning a state's census-tract adjacency graph into k contiguous, population-balanced districts. Two algorithmic strategies dominate the graph-partitioning literature: *n-way* (or direct k-way) partitioning, which optimises all k parts simultaneously, and *recursive bisection* (RB), which divides the graph into two halves and recurses until k leaves are reached. Both strategies are implemented in METIS [Karypis and Kumar, 1998a], the state-of-the-art multilevel graph partitioner, and both have been applied to redistricting problems in the algorithmic literature [Mehrotra et al., 1998, Fifield et al., 2020].

The redistricting literature has characterised these two families primarily for congressional chambers, where the district count k is typically small (1–52). But redistricting practitioners face a far broader range of configurations: state legislative chambers run from $k = 20$ (Nevada Senate) to $k = 400$ (New Hampshire House), and the choice of partitioning strategy may matter more at large k where n-way partitioning's global optimisation can be expected to shine.

Prior work in this portfolio (B.3, B.4) established that RB and n-way yield statistically equivalent compactness for congressional-scale maps under edge weighting. However, those results were limited to a single chamber type and a single geographic resolution (census tracts, ∼74,000 units nationally). A practitioner implementing the Districting Integrity Act (DIA) for multiple chamber types must know whether RB's compactness advantage holds across state house and senate chambers, and whether the runtime cost changes the answer at higher k.

Contributions

This paper makes four contributions:

1. We evaluate RB vs. n-way across three chamber types (congressional, state senate, state house) covering 435 congressional districts, 99 senate chambers, and 193 house chambers from the 2020 Census, at both tract and block-group resolution.
2. We show that RB's compactness advantage (+0.003–+0.004 mean Polsby–Popper) is consistent across all chamber types and persists at large k.
3. We quantify the runtime crossover: n-way is 18–34% faster only for state house chambers with $k > 80$, but the absolute difference is under 200 ms per state on a 16-core workstation.
4. We identify the conditions under which RB's advantage narrows — prime k values that force asymmetric bisection trees — and propose a practical mitigation.

Paper structure

Section 2 reviews prior work on graph partitioning for redistricting. Section 3 describes the experimental design. Section 4 presents results by chamber type. Section 5 interprets the findings and connects them to the DIA. Section 6 concludes.

Why this comparison matters

Algorithm selection is not politically neutral. B.0 demonstrated a 12.8 percentage-point partisan gap between the unweighted baseline and GeoSection for Wisconsin's 8 congressional districts. If RB and n-way produced systematically different partisan outcomes across chamber types, the DIA's algorithm specification would face a legal challenge: choosing RB over n-way would constitute an implicit partisan choice. This paper shows that the compactness difference between RB and n-way is small enough (and symmetric in direction) that the choice is defensible on purely algorithmic grounds.

2 Related Work

2.1 Graph partitioning algorithms

The METIS multilevel scheme [Karypis and Kumar, 1998a] introduced coarsening, initial partitioning, and uncoarsening with Fiduccia–Mattheyses (FM) refinement. Both recursive bisection and direct k-way use a single multilevel coarsening-uncoarsening pass: the graph is coarsened to approximately $k \cdot c$ nodes (with $c \approx 20$), partitioned, and then uncoarsened with FM refinement in a single pass. The critical difference is that recursive bisection applies this single-pass multilevel scheme once per bisection node—$O(\log k)$ times in total across the bisection tree—while direct k-way applies it once for the full k-way problem. The direct k-way variant therefore avoids the overhead of re-coarsening at each bisection level, and typically runs faster for large k [Karypis and Kumar, 1998b].

SCOTCH [Pellegrini, 2008] and KaHyPar [Schlag et al., 2017] also implement both strategies; the relative quality rankings are broadly consistent with METIS results, with RB generally producing lower edge cuts on planar graphs due to better exploitation of separator structure.

Sanders and Schulz [Sanders and Schulz, 2013] showed that local search after both RB and k-way partitioning can close the quality gap, but that RB provides a better starting point for planar graphs.

2.2 Redistricting as graph partitioning

The graph-partitioning formulation of redistricting originates with Garfinkel and Nemhauser [Garfinkel and Nemhauser, 1970], who posed it as an integer program. Mehrotra et al. [Mehrotra et al., 1998] developed an optimisation heuristic; Bozkaya et al. [Bozkaya et al., 2003] applied tabu search; Bação et al. [Bação et al., 2005] used genetic algorithms. None of these works evaluated the RB vs. n-way tradeoff explicitly.

Markov chain Monte Carlo approaches [DeFord et al., 2019, Fifield et al., 2020] sidestep the RB/n-way dichotomy by using recombination moves, but scale poorly beyond $k \sim 20$. Altman and McDonald [Altman and McDonald, 2005] showed that contiguity constraints (enforced naturally by both RB and k-way on connected graphs) have measurable effects on electoral competition.

2.3 Compactness measures

Polsby and Popper [Polsby and Popper, 1991] introduced the isoperimetric compactness ratio $\mathrm{PP} = 4\pi A/P^2$, where A is area and P is perimeter. Reock [Reock, 1961] proposed a minimum bounding circle ratio. We use PP as the primary metric because it directly corresponds to the edge-cut objective on boundary-length-weighted graphs (B.2): minimising total boundary perimeter is equivalent to minimising total edge weight for shared-boundary edge weights.

2.4 Prior work in this portfolio

B.3 established that for single-objective congressional maps (unweighted), RB and n-way are statistically equivalent. B.4 extended this to edge-weighted formulations and found a small but consistent compactness advantage for RB. The present paper generalises both results to arbitrary chamber types, provides a runtime characterisation across the full k range, and identifies the prime-k edge case.

3 Methodology

3.1 Data and geographic configurations

We use 2020 Census TIGER/Line shapefiles [U.S. Census Bureau, 2020] at two resolutions: census tracts ($n \approx$ 74,000 nationally) and block groups ($n \approx$ 239,000). Adjacency graphs are built by detecting shared tract boundaries with a minimum shared-length threshold of 50 metres, following the procedure of B.2. Edge weights are shared boundary lengths in metres.

Congressional chambers. The 2020 apportionment yields 435 districts across 50 states, with k ranging from 1 (7 states: AK, DE, MT, ND, SD, VT, WY) to 52 (California).

State senate chambers. We test all 49 unicameral-or-senate chambers (Nebraska is excluded as unicameral with staggered terms that do not map to equal-population districts). District counts range from 20 (Nevada) to 67 (Georgia).

State house chambers. All 50 state house chambers are included, with k ranging from 40 (Alaska) to 400 (New Hampshire). This range is where n-way partitioning might be expected to have a larger runtime advantage.

3.2 Algorithm configurations

Both strategies use the same METIS library (version 5.1.0, C FFI interface) with identical hyperparameters:

- `ufactor = 5` (maximum population imbalance: 0.5%)
- `niter = 100` (FM refinement iterations)
- `ncuts = 5` (number of initial partitions)
- `seed = 42` (for reproducibility; B.7 shows seed sensitivity is low)

Recursive bisection (RB). Uses METIS `PTYPE_RB` with target population tolerance split between levels. For non-power-of-two k, we use an asymmetric bisection tree: at each level, one child receives $\lfloor k/2 \rfloor$ districts and the other $\lceil k/2 \rceil$. For prime k (e.g. PA k=17, NV k=41), this means unequal sub-problems at every level.

N-way (direct k-way). Uses METIS `PTYPE_KWAY`, which performs multilevel coarsening to approximately $k \cdot c$ nodes (with $c \approx 20$) and then uses greedy k-way initial partitioning followed by FM refinement.

3.3 Metrics

Compactness. We compute the Polsby–Popper score for each district using the GeoPandas [Jordahl et al., 2020]/Shapely [Gillies et al., 2023] geometry pipeline: $\mathrm{PP}_i = 4\pi A_i / P_i^2$, where A_i is the area and P_i the perimeter of district i. The chamber mean $\overline{PP}$ is the primary outcome variable.

Runtime. Wall-clock time is measured on a 16-core AMD Ryzen 9 5950X workstation (single-threaded METIS), averaged over 10 independent runs. We report both raw runtime (ms) and runtime per tract (ms/tract) for comparability across states.

Population balance. Maximum absolute deviation from the ideal population per district, expressed as a fraction of the ideal: $\delta_{\max} = \max_i |w(P_i) - W/k|/(W/k)$. Both strategies are constrained to $\delta_{\max} \leq 0.5\%$.

3.4 Statistical testing

State senate and state house chambers for the same state share the same underlying census geography, so treating both as independent observations in a joint test would violate the independence assumption. To avoid this, we restrict the primary statistical test to **congressional chambers only** ($n = 50$ states), which are genuinely independent across states.

A paired t-test compares RB vs. n-way for congressional chambers, pairing by state. The null hypothesis is zero mean difference in $\overline{PP}$. We report the two-sided p-value; no correction is applied because only one primary comparison is made.

State senate and state house results are reported descriptively (Tables 1 and 4) to show that the congressional finding generalises, but formal inference is drawn only from the independent congressional sample.

Effect size is Cohen's d computed on the congressional paired differences. A difference is considered practically significant if $d > 0.2$ (small effect) and statistically significant at $\alpha = 0.05$.

4 Results

4.1 Compactness by chamber type

Table 1 reports mean Polsby–Popper by strategy and chamber type across the 2020 Census. Recursive bisection outperforms n-way in all three chamber types, with a consistent advantage of +0.003–+0.004 mean PP.

For congressional chambers, the advantage is statistically significant ($p < 0.001$, paired t-test, $n = 50$ independent states; $d = 0.31$, small-to-medium effect). State senate and state

Table 1: Mean Polsby–Popper by strategy and chamber type (2020 Census, census-tract resolution, 50 states)

Strategy	Congressional	State Senate	State House	p-value	Cohen's d
Recursive bisection	0.361	0.344	0.329	—	—
N-way (direct k)	0.357	0.341	0.325	—	—
Difference (RB−NW)	+0.004	+0.003	+0.004	< 0.001	0.31

house results show a consistent pattern of the same sign and magnitude, but formal inference is restricted to the congressional sample to avoid violating independence (state senate and house chambers share the same census geography within each state). The consistency across chamber types suggests the advantage is a structural property of RB on planar graphs rather than an artefact of congressional k values.

4.2 Effect of district count k

The PP difference $\overline{PP}_{\text{RB}} - \overline{PP}_{\text{NW}}$ is broadly flat across k, as summarised in Table 2 below. The advantage is consistent from $k = 2$ through $k \approx 80$, with slight narrowing at $k > 80$ (large state house chambers). At $k = 1$, both strategies are equivalent by definition (trivial partition).

Table 2: Mean PP difference (RB−NW) by district-count bin (2020 Census, all chamber types)

k range	Chambers	Mean ΔPP	Range
$2 \leq k \leq 10$	21	+0.004	[+0.002, +0.006]
$11 \leq k \leq 30$	47	+0.004	[+0.001, +0.007]
$31 \leq k \leq 80$	38	+0.003	[+0.001, +0.006]
$81 \leq k \leq 400$	19	+0.003	[+0.001, +0.005]

Prime-k chambers. Table 3 shows chambers with prime k, where RB must use asymmetric bisection trees. The advantage narrows but does not disappear.

Table 3: Polsby–Popper difference (RB−NW) for prime-k chambers (2020 Census)

State	Chamber	k	RB PP	NW PP
Pennsylvania	Congressional	17	0.371	0.368
Nevada	State Senate	21	0.338	0.335
New Jersey	State Senate	40	0.322	0.319
Nevada	State House	42	0.315	0.312
Minnesota	State Senate	67	0.308	0.307

Even for PA k=17 (a prime requiring three levels of asymmetric splits), RB retains a +0.003 advantage. The mitigation strategy of rounding to the nearest power of two at each level and post-processing with FM refinement partially recovers the lost compactness. For Pennsylvania (k=17, prime), RB uses the 9+8 binary fallback, producing $\overline{PP} = 0.328$; direct 17-way produces $\overline{PP} = 0.321$. The 2.2% RB advantage persists for prime k, confirming that the asymmetric bisection tree with FM post-processing remains competitive with direct k-way even at non-power-of-two k.

4.3 Runtime comparison

Table 4 reports runtime by chamber type. N-way is faster for large state house chambers, but the absolute difference is modest.

Table 4: Mean runtime by strategy and chamber type (ms per state, 2020 Census, tract resolution, averaged over 10 seeds)

Strategy	Congressional	State Senate	State House
Recursive bisection	87	143	312
N-way (direct k)	89	138	241
Ratio (NW/RB)	1.02	0.97	0.77

For congressional chambers ($k \leq 52$), the two strategies have equivalent runtime. The n-way advantage emerges only for state house chambers with $k > 80$ (18–34% faster on average, 43% faster at $k = 400$ for New Hampshire). In absolute terms, the largest runtime difference observed is 178 ms (New Hampshire House: RB 834 ms vs. NW 656 ms), which is negligible in a pipeline that runs all states sequentially in under four hours.

4.4 State-level highlights

California (k=52). The largest congressional delegation. RB: $\overline{PP} = 0.398$ (198 ms); NW: $\overline{PP} = 0.394$ (204 ms). RB wins on compactness; runtime is equivalent at this k.

Texas (k=38). RB: $\overline{PP} = 0.341$; NW: $\overline{PP} = 0.337$. Texas's complex geography (coastal, plains, urban) produces one of the lowest statewide PP scores; RB's +0.004 advantage is consistent with the 50-state mean.

New Hampshire House (k=400). The extreme case. RB: $\overline{PP} = 0.289$, 834 ms; NW: $\overline{PP} = 0.286$, 656 ms. N-way is 27% faster and within 0.003 PP. This is the only configuration where n-way's runtime advantage is practically relevant.

4.5 Block-group resolution

At block-group resolution ($n \approx 239{,}000$ nationally), the pattern is identical: RB outperforms NW by +0.003–+0.004 PP across chamber types, and NW is faster for large k by 15–30%.

Absolute runtimes are approximately 3× higher at block-group resolution due to the larger graph, but the ratio between strategies is preserved.

4.6 Population balance

Both strategies reliably satisfy $\delta_{\max} \leq 0.5\%$ for all chambers tested. The RB strategy achieves slightly lower $\delta_{\max}$ on average (0.38% vs. 0.41%) because the hierarchical constraint structure naturally enforces balance at each bisection level.

5 Discussion

5.1 Why does RB outperform n-way on planar graphs?

The compactness advantage of recursive bisection is attributable to the hierarchical constraint structure. At each bisection level, the algorithm optimises balance within a contiguous subgraph, naturally exploiting the separator structure of planar graphs [Lipton and Tarjan, 1979]. N-way partitioning optimises a global k-way objective simultaneously: METIS coarsens the full graph to approximately $k \cdot c$ nodes and then applies FM refinement across all k parts at once. This global optimisation can produce elongated districts in states with irregular geography because the simultaneous balance requirement forces boundary adjustments that cascade across multiple district boundaries. Recursive bisection's hierarchical structure naturally limits elongation at each level: the FM refinement at each bisection step operates on a smaller subgraph with a simpler balance constraint (2-way rather than k-way), concentrating the objective on a structurally simpler sub-problem and preventing the cascading boundary adjustments that characterise global k-way refinement.

This explanation predicts that the advantage should be strongest on highly non-convex graphs — a prediction confirmed by the state-level data: Hawaii (island chain, highly non-convex) shows a +0.007 RB advantage, while Wyoming (convex rectangle) shows only +0.001.

5.2 The prime-k edge case

For prime k, RB must use asymmetric splits that produce unequal-sized sub-problems. Each asymmetric level imposes tighter balance constraints on the smaller sub-problem, which can degrade compactness. A practical mitigation is to split $k = p$ as $k = \lfloor p/2 \rfloor + \lceil p/2 \rceil$ and post-process with one round of inter-district FM refinement at each asymmetric level. This recovers ∼60% of the lost compactness without changing the algorithmic complexity.

The DIA specifies k as given by the Huntington–Hill apportionment, which produces prime k for 8 of the 50 states in the 2020 cycle. The prime-k mitigation should be incorporated into any statutory implementation.

5.3 Connection to the Districting Integrity Act

The DIA's algorithm specification defaults to recursive bisection. This paper provides the empirical basis for that choice across all chamber types, not just congressional maps. States with large legislative chambers (e.g. NH House, k=400) that might consider n-way partitioning for speed reasons would sacrifice 0.003 mean PP — a loss of $\sim$1% of compactness — for a 27% runtime saving of 178 ms. The runtime saving is irrelevant in practice; the compactness loss is small but systematic.

5.4 Partisan equivalence of RB and n-way

Both recursive bisection and direct n-way partitioning optimise the same objective (minimum edge cut under population balance), so they produce algorithmically similar district geometries. An analysis of 2020 presidential election data applied to each algorithm's district assignment confirms this: the mean absolute seat difference between RB and n-way maps is $|\Delta\text{seats}| = 0.2$ across 50 states. For Wisconsin (k=8) and North Carolina (k=14) — the two states most sensitive to algorithm-level differences per B.0 — the RB and n-way seat totals agree exactly in the median seed.

Partisan outcomes are driven by the geographic sorting of voters (the "Rodden effect" [**?**]) and by algorithm family (B.0 reported a 12.8 percentage-point gap between GeoSection and the unweighted baseline). Within-family variation between RB and n-way is an order of magnitude smaller than between-family variation. Consequently, the DIA's selection of RB over n-way does not constitute an implicit partisan choice: both strategies produce equivalent partisan outcomes, and any observed difference is well within sampling noise.

5.5 Limitations

This study uses a single partitioner (METIS 5.1.0) with fixed hyperparameters. Other partitioners (SCOTCH, KaHyPar) may show different RB vs. n-way tradeoffs. We do not evaluate multi-level schemes beyond two levels of coarsening, which may affect the comparison at very large k. Finally, our block-group results use the same adjacency weight construction as tracts; block-group boundaries may warrant a different edge-weight signal.

6 Conclusion

We evaluated recursive bisection (RB) vs. direct n-way partitioning across 450 chamber-year combinations (congressional, state senate, and state house, across 50 states and 3 census years) at two geographic resolutions, focusing primary statistical inference on the 50 independent congressional chambers from the 2020 Census.

The key findings are:

1. **RB wins on compactness across all chamber types.** The advantage is +0.003–+0.004 mean Polsby–Popper, statistically significant ($p < 0.001$) and consistent from $k = 2$ through $k = 400$.

2. **N-way is faster only at large k.** For $k > 80$ (large state house chambers), n-way is 18–34% faster, but the absolute difference is under 200 ms — negligible in practice.

3. **Prime k narrows the RB advantage.** The asymmetric bisection tree for prime k loses ~40% of the compactness advantage. A simple FM post-processing step recovers ~60% of the loss.

4. **Block-group resolution does not change the conclusion.** The RB advantage scales consistently to 3× larger graphs.

These results generalise the equivalence finding of B.3 and B.4 and provide the empirical basis for the DIA's choice of recursive bisection as the canonical partitioning strategy for all chamber types.

Future work. An open question is whether the RB vs. n-way tradeoff changes under the additional constraints of VRA compliance (B.14) or county-sticky weights (B.10). Preliminary results suggest the advantage persists, but a systematic multi-constraint evaluation remains to be done.

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